Jason Landa

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Education

California State University, Long Beach CA.
Aerospace Engineering (BS), Astronautics Emphasis
Expected graduation date – December 2026
Relevant Coursework: Computational Fluid Dynamics, Thermodynamics, Heat Transfer, Aerodynamics II, Propulsion

Experience

Beach Launch Team (BLT) — CSULB *2025 - Present*

CFD Lead, ANSYS Fluent Instructor

- Conducted 1D/2D CFD modeling of aerodynamic and propulsion-related flows of the current LR101 regenerative cooled engine used in the BLT Theseus rocket.
- Analyzed aerodynamic performance of the rocket skin, modeling boundary-layer growth and transition along the fuselage using varied numerical scheme methods (upwind, central-difference, and higher-order discretization's) to evaluate flow separation and surface heating effects.
- Organized and facilitated ANSYS Fluent training sessions focused on advanced CFD modeling and simulation methodologies.

Propulsion Lead Assistant

- Directed propulsion design efforts toward developing a high-efficiency regeneratively cooled combustion chamber for the next BLT launch vehicle.
- Conducted regenerative cooling performance studies using calibrated infrared thermography, mapping temperature fields across 300+ data points and improving thermal prediction correlation by 20%.
- Executed hydrodynamic flow testing to calculate discharge coefficients across combustion chamber geometry, reducing flow uncertainty by 12% and informing inlet/outlet design adjustments.

Pyrotechnic Igniters Lead

- Directed repeatability and performance testing of pyrotechnic ignition materials, captured and analyzed ignition success rates and thermal response metrics to inform selection of the primary igniter system.
- Implemented engineering redesigns of the pyrotechnic igniter assembly, including revision of the firing circuit, mechanical retention via mounting features, and initiation charge containment to enhance robustness under vibration and thermal loads.
- Developed and implemented standardized procedures for igniter material molding, preparation, and quality verification to ensure consistency and reproducibility across test campaigns.

Gas Dynamics Research Assistant — CSULB *2025 – Present*

Regenerative-cooled rocket engine development – Faculty Advised Research

- Developed two-phase vaporization/heat-transfer models that reduced prediction error by 15%, improving chamber pressure & temperature correlation for chamber design decisions. correlated vaporization behavior with chamber pressure/temperature and informed injector and feed-system design decisions.
- Developed a MATLAB/Python based Prandtl–Meyer Method of Characteristics model to design and optimize a shock-free supersonic nozzle, improving flow uniformity and expansion efficiency, achieving less than 5% flow-angle error for supersonic nozzle design, improving expansion uniformity and theoretical Isp estimates.
- Performed a comprehensive chemical-equilibrium analysis (CEA) trade study across candidate propellant blends, generating carpet (contour/sensitivity) plots to map efficiency and specific-impulse vs. mixture ratio, and recommending optimal propellant/oxidizer combinations for target mission profiles.

AURORA Design Team — Long Beach, CA *2026 – Present*

Propulsion Design Lead

- Conducted Parametric Design Sweep Across multiple pintle injector performance parameters by utilizing cost function analysis to find the most optimal throttleable geometric pintle design choice using MATLAB.
- Leading the propulsion subsystem by surrounding nozzle design and injector design by assigning members tasks which include gas dynamic simulations, Thermomechanical stress analysis, pintle tip spray angle simulations, pintle design sweeps, and combustion stability analysis.

Technical Skills

Software: ANSYS, MATLAB, SolidWorks, Excel, Python, Mastercam, Catiav5,
Laboratory: Wind Tunnels, Pitot-Static Probes, Pressure transducers, hot-wire anemometers.
Skills: CFD (steady/transient turbulent flow, heat transfer), rocket propulsion, workshop instruction, CAD, FEA, CAM toolpath creating, Trade Study Optimization, Lathe, Mill, Wire EDM, Soldering, smoke-wire visualization, schlieren imaging.